

QUEENSTOWN, New Zealand

WMO: 938310

Lat: 45.0175S

Lon: 168.7400E

Elev: 1171

StdP: 14.08

Time Zone: 12.00 (NZL)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	26.4	20.1	15.7	28.5	22.0	17.3	31.2	25.8	40.1	21.8	41.4	6.4	40	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.4	77.8	59.6	75.0	58.7	72.4	57.9	62.5	72.8	61.0	70.3	59.6	68.8	11.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.7	77.0	65.9	57.2	72.8	64.7	55.5	68.5	63.8	28.7	73.0	27.6	70.2	26.7	68.7	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.0	20.1	17.6	DB	22.2	83.2	1.8	2.8	20.9	85.2	19.8	86.9	18.8	88.4	17.5	90.5	(4)
(5)			WB	21.1	65.3	1.8	1.6	19.8	66.5	18.7	67.4	17.7	68.3	16.4	69.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	60.4	60.1	56.1	49.8	45.0	39.7	38.3	41.4	45.9	49.6	53.1	57.4	(6)	
	DBStd	8.96	5.25	5.18	4.86	4.87	5.27	4.89	5.00	4.49	4.65	4.77	5.40	5.09	(7)	
	HDD50	1421	1	1	7	62	171	310	362	269	139	67	27	5	(8)	
	HDD65	5628	160	151	277	456	619	759	827	731	573	478	357	240	(9)	
	CDD50	1303	324	283	196	56	17	1	0	3	16	54	120	233	(10)	
	CDD65	36	17	13	2	0	0	0	0	0	0	0	1	3	(11)	
	CDH74	344	179	104	8	0	0	0	0	0	0	0	9	44	(12)	
	CDH80	35	24	10	0	0	0	0	0	0	0	0	0	1	(13)	
(14) Wind	WSAvg	7.5	8.8	8.3	7.5	6.9	6.3	6.3	6.3	6.5	7.4	8.3	8.9	8.6	(14)	
(15) Precipitation	PrecAvg	32.9	3.1	2.4	2.4	2.4	3.4	3.0	2.5	2.8	2.5	2.8	2.6	2.9	(15)	
	PrecMax	47.9	9.2	5.3	7.0	6.6	6.3	6.3	5.7	11.4	7.2	6.4	10.7	7.0	(16)	
	PrecMin	25.0	0.9	0.5	0.9	0.4	1.2	0.5	0.3	0.2	0.4	1.0	0.0	0.2	(17)	
	PrecStd	5.3	1.5	1.3	1.2	1.6	1.5	1.7	1.2	2.1	1.8	1.3	2.4	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	83.4	81.3	75.1	67.9	62.8	57.0	55.0	58.3	64.3	68.7	75.0	78.2	(19)	
		MCWB	61.3	62.4	58.9	55.0	51.8	47.9	46.0	47.0	49.7	53.4	57.7	58.5	(20)	
	2%	DB	78.5	77.0	71.5	64.3	59.7	53.8	51.9	54.7	60.2	65.1	70.4	74.7	(21)	
		MCWB	59.9	59.9	57.7	53.6	50.7	46.2	44.8	45.8	48.6	52.0	55.8	57.9	(22)	
	5%	DB	75.3	74.3	69.0	61.8	57.1	51.2	49.4	52.2	57.8	62.2	67.1	71.7	(23)	
		MCWB	58.8	59.0	56.9	52.2	49.6	45.2	43.7	44.5	47.6	50.5	54.0	57.1	(24)	
	10%	DB	72.1	71.6	66.4	59.5	54.5	48.9	47.1	50.1	55.5	59.7	64.0	68.5	(25)	
		MCWB	57.9	58.1	55.7	51.2	48.3	44.1	42.3	43.7	46.8	49.3	52.5	55.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.4	65.6	62.6	57.4	54.4	49.7	47.4	48.1	52.0	55.2	60.4	62.5	(27)	
		MCDB	76.3	78.0	69.9	64.7	59.5	54.8	51.6	55.2	59.4	64.6	70.1	72.2	(28)	
	2%	WB	62.3	62.9	60.4	55.1	52.4	47.6	45.8	46.7	50.1	53.2	57.6	60.3	(29)	
		MCDB	73.6	73.0	67.4	61.1	57.2	51.5	50.6	53.0	57.3	62.3	66.3	70.2	(30)	
	5%	WB	60.9	60.8	58.5	53.6	50.5	45.9	44.3	45.3	48.6	51.7	55.6	58.6	(31)	
		MCDB	70.8	69.9	66.3	59.7	55.6	50.3	48.8	51.0	55.8	60.3	64.4	67.7	(32)	
	10%	WB	59.4	59.2	56.9	52.1	48.8	44.3	42.7	44.1	47.4	50.2	53.8	57.2	(33)	
		MCDB	68.9	68.7	64.9	58.3	53.7	48.5	46.7	49.3	54.3	58.1	62.5	66.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	19.5	18.1	16.7	14.3	13.9	14.2	15.1	16.6	17.2	18.0	18.5	(35)	
		MCDBR	26.0	24.6	21.5	19.6	17.4	16.8	16.4	17.6	19.9	21.3	23.1	24.3	(36)	
	5% WB	MCWBR	11.0	10.6	10.7	11.4	11.0	11.4	11.2	10.9	11.4	11.1	11.2	10.7	(37)	
		MCWBR	20.9	21.1	17.9	17.2	15.5	15.2	15.6	17.7	19.1	20.2	20.2	20.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.317	0.311	0.295	0.286	0.277	0.267	0.262	0.279	0.298	0.307	0.309	0.310	(40)	
		taud	2.569	2.605	2.659	2.649	2.629	2.585	2.603	2.604	2.542	2.534	2.560	2.597	(41)	
	Ebn at Noon	Ebn at Noon	319	312	303	282	260	250	263	280	296	310	320	323	(42)	
		Edh at Noon	32	30	25	21	18	16	18	22	28	31	33	32	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1902	1712	1275	843	497	388	458	687	1040	1424	1786	1949	(44)	
	RadStd	RadStd	140	138	75	69	41	43	36	52	95	107	144	141	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)			+0.49	N/A	+1.39	N/A	N/A	N/A	-85	-202	N/A

Nomenclature: See separate page